DNA I

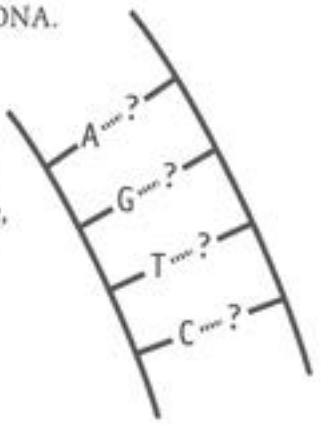
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DNA: the molecular basis of heredity, found in cell nuclei, consisting of a double helix held together by weak bonds between bases which project inward from two chains.

Compared to other molecules, DNA is huge and amazing. However, it is made up of only a few substances: four nitrogen 'bases', a sugar and phosphates. The four bases are adenine (A), thymine (T), cytosine (C) and guanine (G). A base, a sugar and a phosphate form a three-part unit called a nucleotide. The sugar involved is deoxyribose, hence the name deoxyribonucleic acid, or DNA.

Base-pairing

In the same way that jigsaw puzzles fit together, nucleotides have specific pairing rules. Adenine on one strand will pair with thymine on the other, and guanine on one strand will pair with cytosine, producing a ladder-like structure, pulled together by hydrogen bonds into a coil shape, known as a *double helix*.



The diagram on the right shows a short sequence on a DNA 'ladder'.

1 Copy the diagram, using the pairing rules to identify the bases labelled '?'.

2 What do the dotted lines between the bases represent?

Although the amount and proportions of the nitrogen bases vary in all species, the four bases are all present and always follow the base pairing rules: A with T, and G with C.

> Hint: think **G**inger **C**ake. **G**inger and **C**akes go well together, and so do **G** and **C**. If you remember this, that leaves the pairing of A and T remaining by default.

The genetic code can be compared to a four letter alphabet: A, G, C and T. These bases spell the genetic 'words', and these groups of words are known as genes. The language of DNA is a language of proteins. Proteins are made of *polypeptide* chains.

Replication

When DNA replicates (produces a copy of itself), it unwinds and unzips first. The bonds that held the complementary bases together break and each half of the 'zipper' is used as a *template* (model) on which new DNA will be built.

RNA

Like DNA, RNA (ribonucleic acid) is a nucleic acid and controls some of the cell's chemical activities. RNA is formed using DNA as a template and having three of the same bases as DNA. The exception is that instead of thymine it uses uracil (U) to bond with adenine. Combinations of the four 'letters' A, G, C and U then 'spell' the names of amino acids that form the polypeptide chains. For example, the base sequence GCU is the amino acid, alanine. Each 'triplet' of bases is called a *codon*. As the spelling (base sequence) is 'read out' (three letters at a time), the amino acids called for are linked together as a polypeptide chain. Once this is complete the RNA winds itself into specific shapes, depending on the order in which they were assembled, to form organelles, enzymes, and other structures.

3 Give the full names of the bases A, G, C, T and U.

4 Give the full names for DNA and RNA.

5 What substances make up a 'nucleotide'?

6 Complete this sentence: Proteins are made of _____ chains.

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